Quantitative Analysis of Media Bias and Stock Price Dynamics: The 2020 Shock

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***Abstract*—This paper studies how media reporting about large-cap firms changed around the 2020 recession and whether shifts in news stance predict firm-level stock outcomes. Using a large-scale NewsMTSC pipeline to compute daily stance scores from full-text news articles (2015–2025) for S&P 500 firms, we combine NLP-derived stance with firm-day controls and market variables to test structural shifts, causal responses, and predictive linkages. Our identification strategy centers on a pre/post shock design around January 2020, complemented by difference-in-differences, interrupted time series, and time-series VAR exercises. We document the data pipeline, relevance filtering, stance aggregation, and robustness checks used throughout the analysis.**

I. Introduction

The 2020 recession is a natural breakpoint for studying the interaction between media narratives and financial markets. COVID-era coverage expanded in volume and intensity, and a growing body of evidence documents a structural change in news-market dynamics during and after the shock. This project asks a simple, high-level question: did the tone of firm-specific media reporting shift after 2020, and if it did, did markets respond differently to that tone?

We approach the problem with a target-dependent stance framework rather than generic sentiment. NewsMTSC allows us to score whether an article is positive or negative toward a specific firm even when multiple firms are mentioned in the same text. That distinction is central to firm-level testing: a general sentiment score cannot distinguish positive coverage of one firm from negative coverage of another in the same article. Target-dependent stance makes it feasible to construct a daily, firm-level media stance signal and evaluate how it behaves across time and shocks.

The study is grounded in three observations from the project work so far. First, the 2020 shock is widely documented as a period in which media coverage and market behavior changed in a persistent way. Second, the data pipeline needs to be precise: naive keyword search introduces many false positives and inflates noise in the stance signal. Third, if stance is to be linked to returns in a credible way, the design must rule out coverage volume effects, pre-trend differences, and spurious correlations driven by macro shocks.

We therefore build a two-stage pipeline: (i) relevance classification that filters articles to those that actually report on the target firm, and (ii) NewsMTSC stance scoring with confidence thresholds. With this signal we pose three research questions on structural shifts in stance and returns and the post-shock stance-return linkage. Our empirical strategy combines pre/post regressions, interrupted time series, difference-in-differences, VAR and Granger causality, and event studies to triangulate the stance-return channel.

The remainder of the paper proceeds as follows. Section II defines research questions and hypotheses. Section III provides the data story and pipeline. Section IV lays out the methodology and identification strategy. Section V discusses validity threats and robustness. Section VI concludes with planned deliverables.

II. Research Questions and Hypotheses

We formalize three research questions (RQs) that map directly to the project pipeline.

RQ1: Structural shift in media stance. How did media reporting (stance) toward large-cap firms change pre versus post the 2020 recession?
H1: The 2020 shock induced a systematic shift in average daily stance scores after January 2020, conditional on firm fixed effects and controls.

RQ2: Structural shift in returns. Did the same shock change stock return behavior at the firm level?
H2: Mean firm-day returns shifted after 2020, even after controlling for market returns and volatility.

RQ3: Post-shock sensitivity. Did the stance-to-return relationship strengthen after 2020?
H3: The stance-to-return elasticity increased in the post-2020 period for S&P 500 firms.

III. Data and Signal Construction

A. Corpus and Coverage

We begin with a large corpus of scraped full-text news articles (2015–2025). The core sample focuses on large-cap firms that are consistently observable across time, namely the S&P 500. Articles are stored with publication, timestamp, URL, title, first paragraph, and full body text. We de-duplicate on titles and keep a separate record of filtered-out content to preserve traceability.

To avoid severe imbalance across firms and time, we use a stratified design: for each year we compute article counts per firm, take the minimum feasible count across firms, and sample to match that count. This preserves coverage across firms while avoiding a signal driven by a handful of highly covered companies.

B. Sampling and Stratification (Facts and Figures)

The starting panel contains 4,706,589 firm-article links across the top firms and years 2015–2025 (Table I). This raw panel is highly skewed: some firms dominate the coverage while others have sparse years.

We therefore impose two sampling constraints. First, we filter to articles whose titles explicitly contain the company name or ticker and de-duplicate by normalized title per company (title-level deduplication). Second, we select firms that meet a minimum annual coverage threshold: a company is kept if it has at most three years with ≤ 3000 articles; companies failing this criterion are dropped (Table II). This yields a retained set of 22 companies and a stratified, filtered corpus of 483,514 firm-article links (Table III).

Figure 1 shows the long-tail distribution of article counts per company, motivating stratification. Figure 2 summarizes the top-coverage firms in the retained panel.

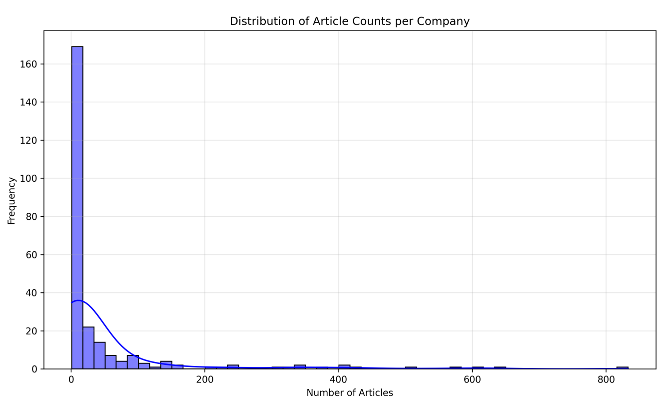

Fig. 1. Distribution of article counts per company. The long tail motivates stratified sampling to avoid coverage-driven bias.

C. Relevance Filtering and Annotation

Keyword matching is not enough for firm targeting (e.g., “Apple” is not always the company). We therefore apply a two-stage relevance pipeline: (i) title-based pre-filtering that retains only articles with the company name in the title (company names, not tickers), and (ii) a supervised relevance classifier trained on manual annotations. The held-out test set for manual annotation is kept completely separate from training.

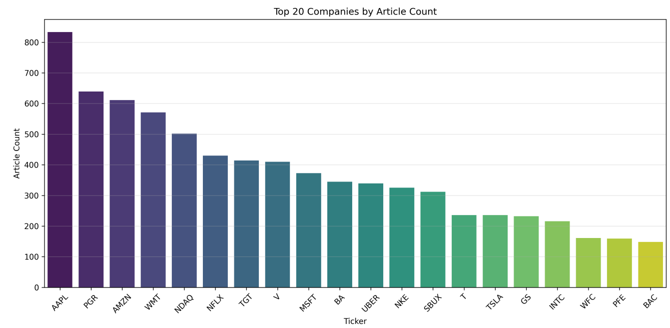

Fig. 2. Top 20 firms by article count in the retained sample.

Our base classifier is DeBERTa-v3 fine-tuned for binary classification using article titles. The first model, trained on 750 labeled articles, achieved high recall but insufficient precision. We then expanded the training data via semi-supervised mining: running the model on a large unlabeled pool and adding only high-confidence predictions ($p_{rel} \geq 0.92$ for relevant, $p_{rel} \leq 0.05$ for irrelevant). This expanded set increases precision and reduces false positives. An ensemble of model variants further improves precision, and the operating threshold is selected via a sweep to meet a target of precision ≥ 0.90.

D. Stance Scoring

Stance is computed using the NewsMTSC model for target-dependent stance. For each qualifying article we obtain probabilities for positive, negative, and neutral stance. The baseline specification retains only high-confidence outputs (prob ¿= 0.9), and we report sensitivity to lower thresholds (0.6 and 0.0). Daily firm-level stance is the average across articles on that day:

$$\text{stance}_{i,t} = \frac{1}{N_{i,t}} \sum_{a \in A_{i,t}} (p_a^+ - p_a^-) \tag{1}$$

where $A_{i,t}$ is the set of high-confidence articles for firm i on day t and $N_{i,t}$ is the count. We also construct article volume and coverage mix covariates to separate tone shifts from coverage shifts.

E. Market Alignment and Controls

The stance series is aligned to firm-day market outcomes: daily returns, S&P 500 index returns, and VIX. We also log-normalize article volume in regressions. This alignment yields the panel used across the RQ1–RQ3 specifications.

IV. Methodology

This section groups the empirical strategy into three layers: (i) structural shifts around the 2020 shock, (ii) stance-return linkage, and (iii) time-series dynamics and event studies.

TABLE I
YEARLY ARTICLE TOTALS IN THE STARTING PANEL (2015–2025)

Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Total articles	570,999	625,582	652,741	525,259	559,027	363,962	291,144	260,115	204,281	204,007	449,472

TABLE II
COVERAGE FILTER FOR FIRM SELECTION

Group	Firms	Rule
Kept	22	At most three years with $\leq$ 3000 articles
Dropped	8	More than three years with $\leq$ 3000 articles

TABLE III
SAMPLE SIZE AFTER FILTERING AND STRATIFICATION

Stage	Total firm-article links
Starting panel	4,706,589
Final stratified panel	483,514

A. RQ1: Stance Shift Around the 2020 Shock

We test for a structural shift in stance using a pre/post specification with firm fixed effects:

$$\text{stance}_{i,t} = \alpha_i + \beta \cdot Post_t + \gamma' X_{i,t} + \varepsilon_{i,t} \tag{2}$$

where $Post_t$ equals 1 after January 2020 (post period: 2020–2025; pre: 2015–2019) and $X_{i,t}$ includes article volume and VIX. If pre-period means are not comparable, we implement Synthetic Control DiD (SCDiD) or use an interrupted time-series specification with level and trend shifts.

B. RQ2: Returns Shift Around the 2020 Shock

The returns analogue mirrors the stance regression, with returns as the outcome and market controls added:

$$\text{return}_{i,t} = \alpha_i + \beta\,\text{Post}_t + \gamma' X_{i,t} + \varepsilon_{i,t} \tag{3}$$

where $X_{i,t}$ includes index returns and VIX to absorb market-wide shocks.

C. RQ3: Stance-Return Linkage

We test the direct stance-return relationship using next-day returns:

$$\text{return}_{i,t+1} = \alpha_i + \beta\,\text{stance}_{i,t} + \gamma' X_{i,t} + \varepsilon_{i,t} \tag{4}$$

To test whether the stance-to-return sensitivity changes post-2020, we use a triple interaction:

$$\begin{aligned}\text{return}_{i,t} = \alpha_i &+ \beta\,\text{Post}_t + \gamma\,\text{stance}_{i,t} \\ &+ \delta(\text{Post}_t \times \text{stance}_{i,t}) + \theta X_{i,t} + \varepsilon_{i,t}\end{aligned} \tag{5}$$

where the interaction $\text{Post}_t \times \text{stance}_{i,t}$ captures post-shock amplification of the stance-return channel.

D. Time-Series Dynamics and Event Studies

We estimate VARs on (stance, returns) to capture bidirectional dynamics and compute impulse response functions. Granger causality tests evaluate whether past stance improves return forecasts beyond past returns, and whether the reverse also holds. We also test for autocorrelation in stance using ARIMA models, which motivates lagged stance controls.

As a non-parametric check, we run event studies around large stance jumps (e.g., ¿2 SD from a rolling mean) and compute abnormal returns in short windows around those events.

E. Robustness and Sensitivity

We perform numerous robustness checks: alternative stance aggregation (median, winsorized means), different confidence thresholds, exclusion of low-coverage firm-days, placebo pre/post cutoffs, and treatments that control for article topical composition. We also test sensitivity to training-test splits for the relevance classifier and manual annotation thresholds.

V. EMPIRICAL METHODOLOGY

We test the project's hypotheses in two escalating stages that follow the logic of the research questions. The first stage asks whether the 2020 shock moved the *level* of two firm-level series—media stance (RQ1) and stock returns (RQ2). This is a static question and is answered with fixed-effects panel regressions. The second stage asks the sharper question behind RQ3: whether the shock changed the *dynamic* relationship between the two series, that is, whether past stance began to forecast returns after the break. This is a time-series question and is answered with a vector autoregression (VAR) and a Granger-causality framework. The two stages require different estimators, which we specify in turn. So that the argument stays readable, the construction and quality auditing of the underlying panel are summarized at a high level here and reported firm-by-firm in the Appendix.

A. Panel Regression Design

The unit of analysis is the firm–day pair (i, t). For each scored article a of firm i published on date t, the NewsMTSC model returns a probability triple (p_a^+, p_a^-, p_a^0) over positive, negative, and neutral stance toward the target firm. We collapse this into a signed article stance $s_a = p_a^+ - p_a^- \in [-1, 1]$, which is large in magnitude only when the model is confident and one-sided, and average it into the firm–day media stance

$$\text{bias}_{it} = \frac{1}{|\mathcal{A}_{it}|} \sum_{a \in \mathcal{A}_{it}} s_a, \tag{6}$$

where $\mathcal{A}_{it}$ is the set of qualifying articles for firm i on day t. The returns outcome is the firm–day simple return $\text{return}_{it} =$

$(P_{it}-P_{i,t-1})/P_{i,t-1}$ from adjusted close prices. The treatment common to both questions is the post-shock indicator $\text{Post}_t = \mathbf{1}[\,t \geq \text{2020-01-01}\,]$, which splits the panel into a pre-period (2015–2019) and a post-period (2020–2025).

The choice of controls follows directly from what could otherwise masquerade as a post-2020 effect. For RQ1 we include firm–day article volume $\text{vol}_{it} = |\mathcal{A}_{it}|$, because averaging a noisy stance signal over more articles mechanically pulls the mean toward zero; holding volume fixed separates a genuine change in tone from a change in coverage. For RQ2 we include the contemporaneous S&P 500 index return sp500_return_t to absorb the market-wide drift and the volatility index vix_t to absorb common volatility regimes; without them the post indicator would merely re-estimate the average market path on each side of 2020 rather than the firm-specific component of interest.

RQ1 is estimated with a two-way fixed-effects specification,

$$\text{bias}_{it} = \alpha_i + \delta_t + \beta\,\text{Post}_t + \gamma\,\text{vol}_{it} + \varepsilon_{it}, \tag{7}$$

where the firm effects α_i absorb every time-invariant firm characteristic and the daily effects δ_t absorb every firm-invariant daily shock. With both present, β identifies only the idiosyncratic within-firm shift in stance after 2020 that is orthogonal to common daily movements. RQ2 mirrors this design with entity (firm) effects and the two market controls,

$$\text{return}_{it} = \alpha_i + \beta\,\text{Post}_t + \gamma_1\,\text{sp500\_return}_t + \gamma_2\,\text{vix}_t + \varepsilon_{it}. \tag{8}$$

Both models are estimated by the within (Mundlak) transformation: each variable is demeaned by its firm and, for RQ1, time means, and OLS is applied to the transformed data, $\hat{\theta} = (\tilde{X}^\top \tilde{X})^{-1}\tilde{X}^\top \tilde{y}$. This is algebraically identical to dummy-variable OLS but avoids storing thousands of fixed-effect columns. Inference uses heteroscedasticity-robust HC1 standard errors, and because firms share common trading-day shocks we additionally report day-clustered standard errors for RQ1, which are robust to arbitrary cross-firm correlation within a trading day and are our preferred uncertainty estimate for the static model.

B. Dynamic Time-Series Design

The regressions above identify shifts in the *level* of each series but are, by construction, silent on the relationship *between* them over time. The question behind RQ3—did the 2020 shock turn media stance into a leading indicator of returns—is inherently dynamic, so we move to a vector autoregression. For a firm with endogenous vector $\mathbf{y}_t = (\Delta\text{bias}_t,\, r_t)^\top$, where Δbias_t is the first difference of the weekly bias index and r_t is the weekly log return, a reduced-form VAR of order p is

$$\mathbf{y}_t = \mathbf{c} + \sum_{\ell=1}^{p} \mathbf{A}_\ell\,\mathbf{y}_{t-\ell} + \mathbf{u}_t, \qquad \mathbf{A}_\ell \in \mathbb{R}^{2\times 2}. \tag{9}$$

The off-diagonal entries of the $\mathbf{A}_\ell$ carry the cross-dynamics: a nonzero $(2,1)$ entry means past stance moves current returns, and a nonzero $(1,2)$ entry means the reverse. The VAR is the minimal model that treats both variables as endogenous and lets the data, rather than an a priori ordering, reveal the direction of predictability.

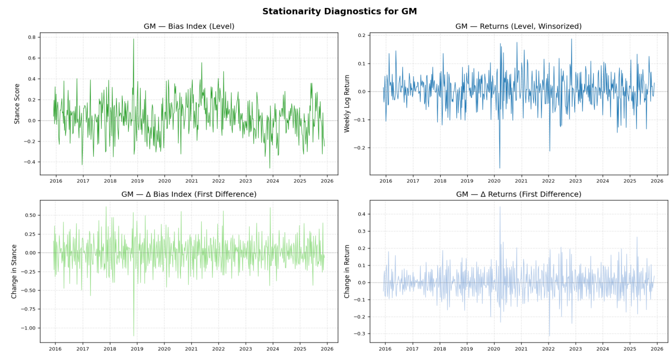

Fig. 3. Stationarity diagnostics for a representative firm (GM): the bias index and weekly return in levels (left) and first differences (right), with the ADF and KPSS verdicts that drive the integration-order classification.

a) From articles to a modeling panel.: Reaching a series that can be fed to Eq. (9) requires several preparatory steps, each of which is audited firm-by-firm in the Appendix and summarized here. A data-inventory pass (Appendix A) shows that daily firm-level stance is too sparse for several firms, so we aggregate to ISO weeks; this yields near-complete coverage (median 99.8% of weeks) for sixteen well-covered firms while retaining roughly 522 weekly observations each. Short coverage gaps are forward-filled and medium gaps linearly interpolated, with long gaps left missing (Appendix B); both an imputed and an observed-only series are kept so that every downstream test can be re-run to confirm results are not artifacts of imputation. Returns are constructed as Monday-anchored weekly log returns and aligned to the bias index (Appendix C).

b) Stationarity dictates the specification.: A VAR in levels is valid only if the series are stationary or jointly cointegrated; otherwise the regression is spurious. Combining the Augmented Dickey–Fuller test (null: unit root) with the KPSS test (null: stationarity), weekly log returns are stationary, $I(0)$, for essentially every firm, whereas the bias index is predominantly $I(1)$ or break-driven in levels but $I(0)$ in first differences (Appendix D, illustrated in Fig. 3). This is why Eq. (9) is specified on $(\Delta\text{bias}_t,\, r_t)$ rather than on the levels. A Johansen trace test finds a cointegrating relation for only one firm, confirming that for the common $(\,I(1), I(0)\,)$ case a differenced VAR, not a VECM, is correct.

c) Lag order and stability.: For each firm the order p is selected by minimizing the Bayesian Information Criterion over $p = 1, \ldots, 8$ (the BIC favors parsimony; a floor of $p = 1$ keeps the model dynamic), giving a modal choice of $p = 3$. Each fitted model is required to be stable—all roots of the companion matrix outside the unit circle—and its residuals are screened for remaining autocorrelation, normality, and ARCH effects (Appendix E). The heavy tails and volatility clustering typical of financial data appear as expected and motivate the robust inference used next.

Fig. 4. Data-driven structural breaks for Visa (V): the return break (2020-03-23) coincides with the COVID crash, while the bias break (2020-09-21) lags it.

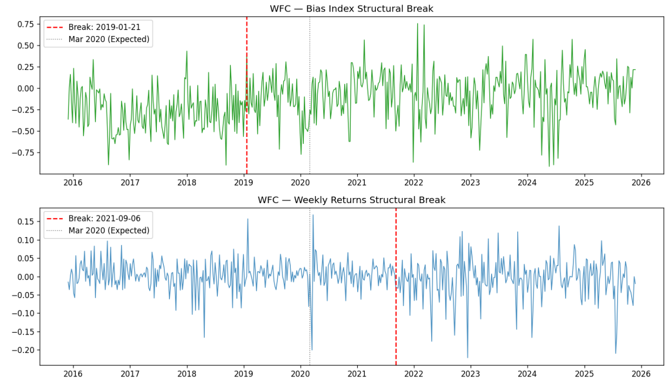

Fig. 5. Structural breaks for Wells Fargo (WFC), the one firm with a cointegrating bias–return relation and significant full-sample causality.

d) Dating the regime change.: Because the central hypothesis concerns a regime change, we date breaks empirically rather than imposing the calendar. A Bai–Perron procedure, solved by dynamic programming for a single mean-shift change point, locates the largest structural break in each weekly series; the bias-index break date defines the pre/post split used for the subsample causality tests (Appendix F, Figs. 4 and 5). Some breaks cluster near the COVID window while others are firm-specific, which is itself evidence that a single imposed 2020 dummy would be too blunt.

e) Granger causality.: Stance Granger-causes returns if the lagged stance coefficients are jointly nonzero in the return equation, i.e. $H_0 : a_1^{(2,1)} = \cdots = a_p^{(2,1)} = 0$ is rejected. We test each direction with a joint Wald (F) test on an OLS regression of each variable on p lags of both, using Newey–West HAC-robust standard errors (`maxlags` $= p$) to neutralize the heteroscedasticity and autocorrelation found above. Firm-by-firm tests are underpowered on a single short series, so we also pool the cross-section into a panel VAR. Unobserved firm heterogeneity is removed with the Arellano–Bover/Helmert transformation (forward mean-differencing),

$$\tilde{y}_{it} = c_{it}\Big(y_{it} - \tfrac{1}{T_{it}} \textstyle\sum_{s>t} y_{is}\Big), \tag{10}$$

which subtracts the mean of all *future* within-firm values and, unlike ordinary demeaning, preserves orthogonality between the transformed variables and the lagged regressors, avoiding the Nickell bias of dynamic fixed-effects panels. The panel model additionally partials out the exogenous controls Post_t, VIX, the S&P 500 weekly log return, and the article count, so that any detected causality cannot be attributed to market-wide volatility or coverage volume. It is estimated by pooled OLS on the transformed, constant-free data with $p = 3$ lags and HAC-robust errors, over the full, pre-break, and post-break samples.

VI. Empirical Results

We report the static results first (RQ1, then RQ2) and then the dynamic results, following the same order in which the methodology was built. Construction and diagnostic tables are deferred to the Appendix; the tables here carry the substantive findings.

TABLE IV
RQ1: Two-Way Fixed-Effects Estimates of Media Stance ($\alpha_i + \delta_t$)

Variable	Coef.	SE (HC1)	p-value
Post	0.014130	0.013623	0.2996
Article Volume	−0.000970	0.000076	< 0.0001
Firm–day obs. $n = 63{,}200$; Firms $= 26$; Days $= 4{,}018$			
Raw articles after filters $= 690{,}586$; Within $R^2 = 0.0017$			

A. RQ1: Media Stance Around the Shock

Table IV reports the two-way fixed-effects estimates. The post coefficient is positive (0.0141) but statistically insignificant ($p = 0.30$), while article volume is strongly negative and highly significant, confirming that heavier coverage is associated with more neutral average tone within a firm–day. Day-clustered inference (SE $= 0.0134$) is almost identical to the HC1 result, so within-day cross-firm dependence does not change the conclusion.

The weak static coefficient is informative rather than a failure of the design. An F-test of the 4,017 daily fixed effects against a firm-only model is jointly significant ($F = 1.135$ versus a 5% critical value of 1.038, $p = 1.1 \times 10^{-8}$), so daily shocks absorb most of the temporal variation in stance. That leaves the coarse stepwise Post_t indicator with little residual variation to explain, which both attenuates β and keeps the within R^2 small. To recover the timing the single step compresses, a monthly event study with firm effects and month-relative dummies around January 2020 (reference month $m = -1$) is reported in Table V: pre-shock coefficients are small and insignificant, consistent with parallel trends, whereas the post-shock effect builds gradually, becomes significant from about three months onward, peaks at $m = +12$ (0.064, $p < 0.001$), and remains elevated at $m = +24$. The 2020 effect is therefore a slow, persistent

TABLE V
RQ1: SELECTED MONTHLY EVENT-STUDY COEFFICIENTS (FIRM FE)

Month	Coef.	SE (HC1)	p-value	Phase
$m = -12$	0.021692	0.018045	0.2293	pre-trend
$m = -6$	0.031757	0.017201	0.0649	pre-trend
$m = 0$	0.031032	0.018354	0.0909	event
$m = +3$	0.042152	0.019168	0.0279	early post
$m = +6$	0.032197	0.018248	0.0777	mid post
$m = +12$	0.064054	0.018797	0.0007	persistent
$m = +24$	0.050394	0.018705	0.0071	long-run

TABLE VI
RQ2: ENTITY FIXED-EFFECTS ESTIMATES OF DAILY RETURNS (HC1 ROBUST SE)

Variable	Coef.	SE (HC1)	p-value
Post	-0.000344	0.000172	0.0451
S&P 500 Return	1.127576	0.010982	< 0.0001
VIX	-0.000001	0.000020	0.9654

uplift rather than a discrete jump, which is precisely why the static term is muted once daily effects are absorbed. The estimator satisfies the Gauss–Markov conditions, with HC1 and clustered errors correcting the heteroscedasticity and large-n asymptotics covering the non-normal residuals; the full diagnostic battery is in Appendix H.

B. RQ2: Returns Around the Shock

The returns panel contains 62,667 firm–day observations across 26 firms (2015-12-01 to 2025-11-24). Table VI reports the entity fixed-effects estimates. Conditional on the index return and VIX, the post-period shift in daily firm returns is negative and marginally significant (-0.000344, $p = 0.045$), about -0.034% per day. The market return loads near unity and highly significantly, as expected for large-cap constituents, while VIX is insignificant once the index return is included. The contrast with RQ1 is the substantive payoff: media stance shows a slow post-2020 uplift while raw returns show only a faint downward mean shift, which motivates asking not whether each series shifts in level but whether the relationship between them changed.

C. Dynamic Linkage: Granger Causality

At the firm level, full-sample causality is sparse, as expected if the relationship is regime-dependent: stance Granger-causes returns for Wells Fargo ($p = 0.023$) and weakly for Morgan Stanley ($p = 0.058$), and the reverse direction is significant for Goldman Sachs ($p = 0.012$); the remaining firms show no significant direction (Appendix G). The informative test splits each firm at its empirical bias break. Several firms that are quiet in the full sample show post-break causality absent pre-break—Wells Fargo (bias→returns $p = 0.013$ post versus 0.345 pre) and Morgan Stanley (bias→returns $p = 0.052$ post) are the clearest—with feedback emerging post-break for Boeing, Goldman Sachs, and Morgan Stanley. This pre/post asymmetry is the firm-level analogue of the structural shift

TABLE VII
PANEL VAR GRANGER CAUSALITY (15 FIRMS, HELMERT-TRANSFORMED, HAC)

Sample	N	Bias→Ret	Ret→Bias
Full	7493	0.976	0.200
Pre-break	4168	0.245	0.801
Post-break	3265	0.617	0.187

the regressions probe: the link is not constant but switches on after the break for a subset of firms.

Pooling the cross-section into the panel VAR buys statistical power, yet Table VII shows that neither Granger direction is significant in any subsample once VIX, the index return, and article volume are controlled.

D. Synthesis

The three results compose a coherent picture. Media stance toward large-cap firms did shift after 2020, but as a gradual uplift rather than a discrete jump; firm returns shifted only faintly once market factors are removed; and the dynamic link between stance and returns, where it appears at all, switches on after the firm-specific break for a handful of firms and does not aggregate into a market-wide channel. The panel VAR is exactly the test that would manufacture a spurious aggregate effect if the controls were omitted, and it declines to do so. The honest conclusion is that the post-2020 media–market relationship is real but heterogeneous and firm-specific, not a uniform regularity—a finding that the staged design, from static panels to a controlled panel VAR, is built to deliver credibly. We collect the construction and diagnostic detail underlying every stage in the Appendix and discuss residual threats to this interpretation next.

VII. IDENTIFICATION, VALIDITY, AND ROBUSTNESS

We address three main threats. First, macro shocks may move both coverage and prices; we control with index returns and VIX. Second, article volume differs widely across firms; we include volume covariates and run trimmed samples. Third, parallel trends may not hold; we use pre-trend checks, placebo cutoffs, and SCDiD when needed.

Robustness checks include alternate stance thresholds (0.9, 0.6, 0.0), alternative aggregation (mean vs median), lagged stance controls, sector controls, and quantile regressions to test for tail effects in returns.

VIII. CONCLUSION AND NEXT STEPS

This draft provides a full narrative of the project to date: why the 2020 shock matters, how we build a target-dependent stance signal, and how we test for structural changes and market effects. Next steps include finalizing the firm panel, completing the relevance classifier evaluation, computing the daily stance series, and running the full RQ1–RQ3 specifications with robustness tables and figures.

TABLE VIII
PER-FIRM COVERAGE IN THE MODELING PANEL

Ticker	Articles	Daily Cov. (%)	Zero Days (%)	Weekly Cov. (%)
ABNB	13,771	116.3	17.1	99.8
AMZN	50,366	135.4	3.3	100.2
T	139,701	139.6	1.5	99.0
BA	47,140	133.1	4.9	100.2
BAC	7,976	95.7	31.7	99.8
GM	22,914	126.2	9.9	100.2
GS	16,775	118.8	15.2	100.0
INTC	22,056	124.6	11.0	100.2
MCD	26,849	127.9	8.6	100.2
MSFT	52,245	136.0	2.9	100.2
MS	54,487	139.0	0.7	100.2
SBUX	19,955	119.4	15.8	99.0
UBER	36,774	125.1	11.7	99.0
V	77,290	138.8	2.1	98.6
WMT	38,196	133.7	5.7	99.0
WFC	8,774	78.4	44.7	97.2

TABLE IX
WEEKLY GAP CLASSIFICATION AND IMPUTATION COUNTS

Ticker	Cov. (%)	Gaps A	Gaps B	Longest	Imputed
ABNB	99.7	1	0	2	2
AMZN	100.0	0	0	0	0
T	98.8	0	1	7	7
BA	100.0	0	0	0	0
BAC	99.7	2	0	1	2
GM	100.0	0	0	0	0
GS	99.8	1	0	1	1
INTC	100.0	0	0	0	0
MCD	100.0	0	0	0	0
MSFT	100.0	0	0	0	0
MS	100.0	0	0	0	0
SBUX	98.8	0	1	7	7
UBER	98.8	0	1	7	7
V	98.4	1	1	7	9
WMT	98.8	0	1	7	7
WFC	97.2	9	1	7	16

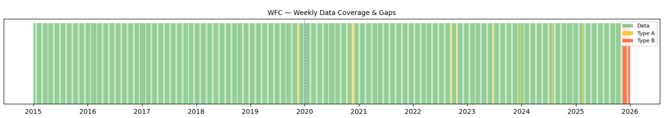

Fig. 6. Weekly coverage and gap map for Wells Fargo (WFC), the lowest-coverage retained firm; it still clears the $\geq 60\%$ threshold after imputation.

APPENDIX A
DATA INVENTORY AND COVERAGE

Table VIII reports per-firm article coverage for the sixteen firms carried into the weekly modeling panel. Daily coverage exceeds 100% for most firms (more than one article per trading day on average); the share of zero-article days motivates the move to a weekly frequency, at which coverage is near-complete. Netflix was audited but dropped at the gap stage (weekly coverage 59.2%); the firm-level inventory of sparse names excluded at this stage (e.g. MMM, AAPL, ETN, NDAQ, NKE, ORCL, PGR, TSLA, each below 30% daily coverage) is omitted for brevity.

APPENDIX B
GAP ANALYSIS AND IMPUTATION

Weekly gaps are classified by length: Type A (1–2 weeks) are forward-filled and Type B (3–8 weeks) linearly interpolated, while Type C (> 8 weeks) are left missing. Across all firms only 58 observations are imputed (16 forward-fill, 42 interpolation), and every firm clears the $\geq 60\%$ coverage threshold with no gap exceeding eight weeks (Table IX). The coverage map for the sparsest retained firm is shown in Fig. 6.

APPENDIX C
RETURNS CONSTRUCTION AND WINSORIZATION

Weekly log returns are computed from Monday-anchored ISO weeks and aligned to the bias index (Table X). Extreme weekly moves ($> \pm 15\%$) cluster in the March 2020 crash window. Returns are winsorized at ± 5 standard deviations from each firm's mean, with AT&T (T) excluded from winsorization so that its outlier behavior can be compared against the others.

APPENDIX D
STATIONARITY AND COINTEGRATION

Integration orders combine the ADF and KPSS verdicts on the levels and first differences of each series (Table XI). Returns are $I(0)$ almost everywhere; the bias index is predominantly $I(1)$ or break-driven in levels. A Johansen trace test detects cointegration only for Wells Fargo (trace $= 340.20$ at $r = 0$ versus a 95% critical value of 15.49), which is the single VECM candidate; for the common $(\, I(1), I(0)\,)$ pairing cointegration is impossible, confirming the differenced VAR.

APPENDIX E
VAR LAG SELECTION AND RESIDUAL DIAGNOSTICS

Lag orders are chosen by BIC over $p = 1, \ldots, 8$; Table XII reports the AIC and BIC choices, the selected order, and residual diagnostics. Every selected model is stable (minimum companion root > 1). The Portmanteau test confirms adequate lag length for most firms, Jarque–Bera rejects normality everywhere (heavy tails), and Engle's ARCH test flags volatility clustering in several return residuals—together motivating the HAC-robust causality tests. WMT is excluded from this specification pass.

APPENDIX F
STRUCTURAL BREAK DATES

Bai–Perron single-break dates for each weekly series, with the resulting pre/post observation counts, are reported in Table XIII. The bias-index break defines the subsample split for the causality tests. Examples are plotted in Figs. 4 and 5.

APPENDIX G
FIRM-LEVEL GRANGER CAUSALITY

Table XIV reports HAC (p-value) Granger tests for every firm, in both directions, over the full sample and the pre/post

TABLE X
WEEKLY LOG-RETURN QUALITY AND WINSORIZATION (16 FIRMS)

Ticker	Weeks	Mean	Std	Min	Max	Extreme ($>\pm 15\%$)	Winsorized
ABNB	522	0.00265	0.03562	−0.1443	0.1399	0	0
AMZN	522	0.00235	0.03230	−0.1829	0.1138	1	1
T	522	0.00147	0.06861	−0.4093	0.3331	18	skipped
BA	522	−0.00046	0.03924	−0.2849	0.1939	4	1
BAC	260	−0.00090	0.06274	−0.1815	0.1896	5	0
GM	522	0.00503	0.05559	−0.2923	0.1868	6	1
GS	522	0.00298	0.03935	−0.2026	0.1322	2	1
INTC	522	0.00224	0.03584	−0.1571	0.1397	1	0
MCD	522	0.00251	0.04381	−0.2056	0.2160	5	0
MSFT	522	0.00209	0.04499	−0.2197	0.1940	6	0
MS	522	0.00335	0.04431	−0.2955	0.2533	6	2
SBUX	522	0.00264	0.04417	−0.1883	0.1934	3	0
UBER	522	0.00250	0.04123	−0.2416	0.2012	4	1
V	522	0.00154	0.05526	−0.2859	0.1954	6	1
WMT	522	0.00154	0.05526	−0.2859	0.1954	6	1
WFC	522	0.00012	0.04667	−0.2215	0.1680	10	0

TABLE XI
INTEGRATION ORDER OF BIAS AND RETURN SERIES

Ticker	(Bias, Return)	Note
ABNB	$(I(1), I(0))$	bias break-driven
AMZN	$(I(1), I(0))$	
T	$(I(1), I(0))$	
BA	$(I(0), I(1))$	return break-driven
BAC	$(I(1), I(0))$	
GM	$(I(0), I(0))$	both stationary
GS	$(I(1), I(0))$	
INTC	$(I(1), I(0))$	
MCD	$(I(1), I(0))$	
MSFT	$(I(1), I(0))$	
MS	$(I(1), I(0))$	
SBUX	$(I(1), I(0))$	
UBER	$(I(1), I(0))$	
V	$(I(1), I(0))$	
WMT	$(I(1), I(0))$	
WFC	$(I(1), I(1))$	cointegrated (VECM)

subsamples defined by the bias break. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

APPENDIX H
RQ1 REGRESSION DIAGNOSTICS

Table XV summarizes the classical-linear-model diagnostics for the RQ1 two-way fixed-effects estimator. Orthogonality and conditioning are satisfied; heteroscedasticity and non-normality are present and handled by the HC1/clustered errors and large-sample asymptotics, respectively; collinearity is negligible.

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TABLE XII
VAR LAG SELECTION AND RESIDUAL DIAGNOSTICS (15 FIRMS)

Ticker	AIC	BIC	Chosen	N	Stable	Min Root	Portm. p	JB p	ARCH(B) p	ARCH(R) p
ABNB	6	3	3	521	✓	1.56	0.0553	0.0000	0.0264	0.0301
AMZN	7	2	2	521	✓	1.99	0.0006	0.0000	0.1811	0.0000
T	6	4	4	521	✓	1.40	0.0985	0.0000	0.3270	0.8143
BA	4	4	4	521	✓	1.37	0.3284	0.0000	0.0784	0.0000
BAC	4	1	1	259	✓	2.02	0.0163	0.0000	0.0640	0.2924
GM	7	3	3	521	✓	1.55	0.0137	0.0000	0.0000	0.0003
GS	8	5	5	521	✓	1.28	0.0410	0.0000	0.5783	0.0007
INTC	8	2	2	521	✓	1.80	0.0008	0.0000	0.1713	0.0000
MCD	8	3	3	521	✓	1.56	0.0003	0.0000	0.0002	0.0000
MSFT	8	3	3	521	✓	1.65	0.0115	0.0000	0.0582	0.2155
MS	7	3	3	521	✓	1.53	0.0001	0.0000	0.6324	0.0000
SBUX	8	2	2	521	✓	1.76	0.0035	0.0000	0.0378	0.0000
UBER	8	3	3	521	✓	1.53	0.0220	0.0000	0.0000	0.0000
V	8	3	3	521	✓	1.65	0.0005	0.0000	0.0002	0.0000
WFC	6	5	5	521	✓	1.28	0.4343	0.0000	0.0340	0.0000

TABLE XIII
EMPIRICAL STRUCTURAL BREAK DATES

Ticker	Bias Break	Return Break	Pre	Post
ABNB	2022-05-23	2021-09-06	338	184
AMZN	2018-02-05	2022-01-03	114	408
T	2022-05-02	2021-09-27	335	187
BA	2024-01-08	2022-01-03	423	99
BAC	2022-04-25	2022-12-26	72	188
GM	2022-10-10	2017-10-30	358	164
GS	2019-05-13	2021-11-22	180	342
INTC	2024-03-25	2023-03-20	434	88
MCD	2022-03-28	2023-10-30	330	192
MSFT	2023-06-12	2021-09-27	393	129
MS	2023-12-11	2023-10-30	419	103
SBUX	2021-11-08	2022-10-03	310	212
UBER	2018-04-02	2024-04-01	122	400
V	2020-09-21	2020-03-23	251	271
WMT	2024-07-22	2020-03-23	451	71
WFC	2019-01-21	2021-09-06	164	358

TABLE XIV
FIRM-LEVEL GRANGER CAUSALITY, FULL AND SUBSAMPLE (HAC p-VALUES)

		Full		Pre-break		Post-break	
Ticker	Lag	B→R	R→B	B→R	R→B	B→R	R→B
ABNB	3	0.992	0.877	0.667	0.690	0.629	0.623
AMZN	2	0.732	0.759	0.196	0.339	0.815	0.278
T	4	0.947	0.345	0.207	0.571	0.122	0.348
BA	4	0.232	0.295	0.196	0.775	0.543	0.014**
BAC	1	0.845	0.985	0.836	0.862	0.676	0.778
GM	3	0.218	0.765	0.304	0.489	0.451	0.835
GS	5	0.757	0.012**	0.699	0.833	0.902	0.007***
INTC	2	0.872	0.305	0.479	0.158	0.173	0.696
MCD	3	0.192	0.258	0.259	0.027**	0.077*	0.484
MSFT	3	0.680	0.134	0.137	0.234	0.309	0.477
MS	3	0.058*	0.193	0.119	0.313	0.052*	0.016**
SBUX	2	0.971	0.379	0.328	0.582	0.243	0.414
UBER	3	0.750	0.192	0.489	0.980	0.856	0.190
V	3	0.981	0.174	0.873	0.157	0.827	0.629
WFC	5	0.023**	0.307	0.345	0.247	0.013**	0.636

TABLE XV
RQ1 SUMMARY DIAGNOSTICS

Test	Statistic	Implication
Residual mean	-9.0×10^{-19}	orthogonality OK
Condition number	157.46	numerically stable
Breusch–Pagan (LM)	190.25	heteroscedastic (HC1 used)
Jarque–Bera	4541.18	fat tails (n large)
VIF (both regressors)	≈ 1.00	no collinearity
Max pairwise corr.	0.0189	negligible
Durbin–Watson	1.673	mild autocorrelation